TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Substitution: get a variable alone, put its expression into the other equation, solve, substitute back.
Variable drops out? False statement (like 0 = 7): no solution. True statement (like 5 = 5): infinite.
Check: put both values into both equations.

1. Solve by substitution: $y = 3x$ and $x + y = 20$.

Tip: Replace y with 3x.

A. (15, 5)

B. (4, 12)

C. (5, 15)

2. Which equation do you get by substituting $y = 2x - 3$ into $4x + y = 15$?

Tip: Put the whole expression in place of y.

F. $4(2x - 3) + y = 15$

G. $4x + 2x = 15$

H. $4x + 2x - 3 = 15$

3. Solve by substitution: $y = x + 4$ and $3x + y = 24$.

> **Tip:** Replace y with x + 4.

3x + (_____) = 24

4x + 4 = 24, so x = _____

y = _____ + 4 = _____